

Project Problem Statement

Image Authentication via Benign-Invariant Feature Mapping

Tushar Yadav

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Problem Definition

The core objective of this project is to develop a mapping function $F : I \rightarrow Y$ that maps any input image I to a compact descriptor $Y = F(I)$. This descriptor must enable robust authentication by distinguishing between two specific classes of image variations:

- **Benign Variants (I_k):** Images subjected to acceptable transformations such as JPEG recompression, denoising, or brightness adjustments[cite: 7].
- **Malicious Variants (I_m):** Images subjected to content-altering manipulations, including deepfakes, splicing, or semantic object removal.

Mathematical Constraints

To achieve robust authentication, the function F and a distance metric d_Y must satisfy two conflicting requirements:

1. **Benign Invariance:** The distance between the descriptor of the original image I_0 and any benign variant I_k must be minimal:

$$d_Y(F(I_0), F(I_k)) < \epsilon \quad (1)$$

where ϵ is a small threshold tolerance.

2. **Semantic Sensitivity:** The distance between the descriptor of the original image I_0 and any semantically manipulated version I_m must be significant:

$$d_Y(F(I_0), F(I_m)) \geq \delta \quad (2)$$

where $\delta \gg \epsilon$.